

Variables are simply names that refer to objects. *Attributes* are, to quote the official documentation, “any name following a dot”

(<https://docs.python.org/3/tutorial/classes.html#python-scopes-and-namespaces>). Attributes are associated with objects (the name before the dot/period). For example, enter the following into the interactive shell:

```
>>> import datetime
>>> spam = datetime.datetime.now()
>>> spam.year
2018
>>> spam.month
1
```

In this code example, `spam` is a variable that contains a `datetime` object (returned from `datetime.datetime.now()`), and `year` and `month` are attributes of that object. Even in the case of, say, `sys.exit()`, the `exit()` function is considered an attribute of the `sys` module object.

Other languages call attributes *fields*, *properties*, or *member variables*.

Function vs. Method

A *function* is a collection of code that runs when called. A *method* is a function (or a *callable*, described in the next section) that is associated with a class, just as an attribute is a variable associated with an object. Functions include built-in functions or functions associated with a module. For example, enter the following into the interactive shell:

```
>>> len('Hello')
5
>>> 'Hello'.upper()
'HELLO'
>>> import math
>>> math.sqrt(25)
5.0
```

In this example, `len()` is a function and `upper()` is a string method. Methods are also considered attributes of the objects they’re associated with. Note that a period doesn’t necessarily mean you’re dealing with a method instead